

CSA0501
DATABASE MANAGEMENT SYSTEMS
LAB EXPERIMENT

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EXP 2:

AIM:

To add constraints such as PRIMARY KEY, FOREIGN KEY, UNIQUE KEY, and CHECK to database tables using DDL commands (ALTER TABLE) in MySQL, and to verify their enforcement by inserting valid and invalid data.

COMMANDS:(PRIMARY, FOREIGN KEY, UNIQUE, CHECK)

ALGORITHM:

1. Create a database, select it using USE, and create the base tables (STUDENT2, STUDENT3, FACULTY, DEPARTMENT, COURSE) without any constraints.
2. Add a PRIMARY KEY constraint to STUDENT2(RegNo) using ALTER TABLE, insert valid records, then attempt a duplicate RegNo to verify rejection.
3. Add a PRIMARY KEY constraint to FACULTY(FacNo) and a CHECK constraint on Gender ('M' or 'F'); insert valid values, then an invalid value to verify rejection.
4. Add a PRIMARY KEY constraint to DEPARTMENT(DeptNo) and insert sample records.
5. Add a PRIMARY KEY constraint to COURSE(CourseNo) and a CHECK constraint on SemNo (1 to 6); insert a valid SemNo, then an invalid SemNo to verify rejection.
6. Add a UNIQUE constraint to STUDENT2(S1Name); insert a unique name, then a duplicate name to verify rejection.
7. Add a FOREIGN KEY constraint to STUDENT3(RegNo) referencing STUDENT2(RegNo) with ON DELETE CASCADE and ON UPDATE CASCADE.
8. Insert valid RegNo values into STUDENT3, then attempt a RegNo not present in STUDENT2 to verify rejection.
9. Delete a record from STUDENT2 and verify that the corresponding record in STUDENT3 is automatically removed (cascade effect).
10. Use DESCRIBE, SHOW CREATE TABLE, and information_schema.TABLE_CONSTRAINTS to verify all applied constraints, then stop.

QUESTIONS:

1) Alter the table STUDENT2 with following structure.

```
mysql> INSERT INTO STUDENT2 VALUES (101, 'Nikhil', 'AI & DS');
Query OK, 1 row affected (0.02 sec)

mysql> INSERT INTO STUDENT2 VALUES (102, 'Ravi', 'CSE');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO STUDENT2 VALUES (101, 'Kumar', 'ECE');
ERROR 1062 (23000): Duplicate entry '101' for key 'student2.PRIMARY'
mysql> ALTER TABLE FACULTY
    -> ADD PRIMARY KEY (FacNo);
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

2) Alter the table name FACULTY with following structure.

```
mysql> ALTER TABLE FACULTY
    -> ADD CONSTRAINT chk_gender CHECK (Gender IN ('M', 'F'));
Query OK, 0 rows affected (0.14 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> INSERT INTO FACULTY VALUES (1, 'Dr. Suresh', 'M');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO FACULTY VALUES (2, 'Dr. Priya', 'F');
Query OK, 1 row affected (0.01 sec)
```

3) Alter the table name DEPARTMENT with following structure.

```
mysql> ALTER TABLE DEPARTMENT
    -> ADD PRIMARY KEY (DeptNo);
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> INSERT INTO DEPARTMENT VALUES (1, 'AI & DS');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO DEPARTMENT VALUES (2, 'CSE');
Query OK, 1 row affected (0.01 sec)
```

4) Alter the table name **COURSE** with following structure.

```
mysql> INSERT INTO COURSE VALUES (501, 'DBMS', 5);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO COURSE VALUES (502, 'DAA', 4);
Query OK, 1 row affected (0.01 sec)
```

5) Alter the table name **STUDENT2** with following structure.

```
mysql> ALTER TABLE STUDENT2
    -> ADD CONSTRAINT uq_s1name UNIQUE (S1Name);
Query OK, 0 rows affected (0.09 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> INSERT INTO STUDENT2 VALUES (103, 'Meena', 'IT');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO STUDENT2 VALUES (104, 'Nikhil', 'ECE');
ERROR 1062 (23000): Duplicate entry 'Nikhil' for key 'student2.uq_s1name'
```

6) After the **STUDENT2** & **STUDENT3** tables are successfully created, test if you can add a constraint **FOREIGN KEY** to the **RegNo** of this table.

```
mysql> INSERT INTO STUDENT2 VALUES (104, 'Nikhil', 'ECE');
ERROR 1062 (23000): Duplicate entry 'Nikhil' for key 'student2.uq_s1name'
mysql> ALTER TABLE STUDENT3
    -> ADD CONSTRAINT fk_regno
    -> FOREIGN KEY (RegNo) REFERENCES STUDENT2(RegNo)
    -> ON DELETE CASCADE
    -> ON UPDATE CASCADE;
Query OK, 0 rows affected (0.15 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> INSERT INTO STUDENT3 VALUES (101, 92);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO STUDENT3 VALUES (102, 85);
Query OK, 1 row affected (0.01 sec)
```

OUTPUTS:

```
mysql> SELECT * FROM STUDENT3; -- the row with RegNo 102 should be gone
+-----+-----+
| RegNo | Attendance |
+-----+-----+
| 101   | 92         |
+-----+-----+
1 row in set (0.01 sec)
```

```
mysql> -- See full table structure with constraints
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> DESCRIBE STUDENT2;
```

Field	Type	Null	Key	Default	Extra
RegNo	int	NO	PRI	NULL	
S1Name	varchar(30)	YES	UNI	NULL	
Branch	varchar(20)	YES		NULL	

3 rows in set (0.06 sec)

```
mysql> SELECT CONSTRAINT_NAME, CONSTRAINT_TYPE
-> FROM information_schema.TABLE_CONSTRAINTS
-> WHERE TABLE_SCHEMA = 'csa05_lab' AND TABLE_NAME = 'STUDENT2';SELECT CONSTRAINT_NAME, CONSTRAINT_TYPE
```

CONSTRAINT_NAME	CONSTRAINT_TYPE
PRIMARY	PRIMARY KEY
uq_s1name	UNIQUE

2 rows in set (0.03 sec)

Result: DDL Commands with Primary, Foreign, Unique, Check constraints are updated and verified.